HAONAN SI

Curriculum Vitae

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Google Scholar

ACADEMIC PROFILE

Haonan Si is a Postdoctoral Researcher, Shuimu Scholar, and Zhang Keqian Named Postdoctoral Fellow in the Department of Electronic Engineering at Tsinghua University, working with Prof. Yuan Shen since August 2026. He received his Ph.D. in Information and Communication Engineering from the University of Electronic Science and Technology of China (UESTC) in 2026, under the supervision of Prof. Xiansheng Guo. From November 2024 to October 2025, he was a visiting Ph.D. researcher at Queen Mary University of London (QMUL), supervised by Prof. Yuanwei Liu and Dr. Jin Zhang. His research focuses on indoor localization and intelligent sensing, with particular interests in data-driven positioning, learning-based sensing, and robust localization under heterogeneous environments.

- Publications: Multiple first-author journal papers in IEEE TWC, TCOM, TVT, IoT-J, TCCN, and IEEE Wireless Communications.
- Citations: 504 Google Scholar citations, h-index 14 (user-supplied Scholar snapshot, Oct. 2026).
- Intellectual property: 6 granted Chinese invention patents.

RESEARCH INTERESTS

Indoor localization; wireless sensing; crowdsourcing and trajectory-driven learning; reinforcement learning for positioning; robust learning under domain shift; intelligent transportation (AVP/parking digital twins); array sensing (DOA/CRLB).

EDUCATION

University of Electronic Science and Technology of China (UESTC), Chengdu, China
Ph.D. (Integrated M.S.–Ph.D.) in Information and Communication Engineering 2021–2026

- Graduate ranking: Top 5% (program cohort).
- Advisor: Prof. Xiansheng Guo.
- Co-advisors/collaborators: Prof. Nirwan Ansari (NJIT, IEEE Fellow); Prof. Dusit Niyato (NTU, IEEE Fellow).
- Selected honors & scholarships: Tencent Scholarship (Special Class/Top Award), China National Scholarship, CSC Scholarship, UESTC First-Class Academic Scholarship, Outstanding Graduate Student, etc.

North University of China, Taiyuan, China
B.Eng. in Communication Engineering Jun. 2021

- Recommended for postgraduate study; comprehensive ranking: 2/147.
- Selected honors: Mathematical Contest in Modeling (MCM/ICM), Meritorious Winner (M Award); Individual Scholarship (High-Level Academic Paper).
- Collaborating advisor: Prof. Xingling Shao (Vice Dean, School of Electrical and Control Engineering; Top 2% Scientists Worldwide).

APPOINTMENTS & RESEARCH EXPERIENCE

Postdoctoral Researcher, Tsinghua University, Beijing, China Aug. 2026–Present
Department of Electronic Engineering

- Shuimu Scholar; Zhang Keqian Named Postdoctoral Fellow.
- Postdoctoral advisor: Prof. Yuan Shen.
- Research: localization, sensing, and networked intelligence.

Visiting Ph.D. Researcher Nov. 2024–Oct. 2025
Queen Mary University of London (QMUL), London, UK
Department of Electronic Engineering

- Supervisors: Prof. Yuanwei Liu and Dr. Jin Zhang.
- Research: communications engineering and array sensing (e.g., DOA/CRLB with continuous aperture arrays).

Algorithm R&D Intern, Chengdu Fangmi Technology, Chengdu, China Jun. 2023–Jun. 2024

- Algorithm development and engineering implementation (details available upon request / NDA permitting).

Algorithm R&D Intern, Sichuan Daodaojiao Technology, China Jun. 2024–Nov. 2024

- Applied algorithm R&D in industrial settings (details available upon request / NDA permitting).

PUBLICATIONS

Journal Articles (Published / Accepted)

First Author

1. **H. Si**, Z. Wang, X. Guo, J. Zhang, Y. Liu, "Joint DOA and Attitude Sensing Based on Tri-Polarized Continuous Aperture Array," *IEEE Transactions on Wireless Communications*, accepted for publication, 2026.
2. **H. Si**, Y. Liu, X. Guo, G. O. Boateng, N. Ansari, Y. Wang, "Spatial and Temporal Multi-Agent Interactive Localization: A Lightweight Personalized Federated Learning Approach," *IEEE Transactions on Communications*, accepted for publication, 2026.
3. **H. Si**, Z. Wang, X. Guo, J. Zhang, Y. Liu, "DOA Estimation via Continuous Aperture Arrays: MUSIC and CRLB," *IEEE Transactions on Wireless Communications*, 2026.
4. **H. Si**, X. Guo, G. O. Boateng, H. Xia, "Cross-Scenario Foundation Localization Models: Architecture, Key Technologies, and Challenges," *IEEE Wireless Communications*, 2025.
5. **H. Si**, X. Hou, J. Wang, G. O. Boateng, Z. Zhang, X. Guo, D. Niyato, "Unsupervised Localization Toward Crowdsourced Trajectory Data: A Deep Reinforcement Learning Approach," *IEEE Transactions on Wireless Communications*, vol. 24, no. 9, pp. 7970–7984, 2025.
6. **H. Si**, X. Guo, G. O. Boateng, Y. Yang, N. Ansari, "Spatial Feature Topology-Based Heterogeneous Knowledge Transfer Framework for Long-Term Fingerprint Positioning," *IEEE Transactions on Vehicular Technology*, 2025.
7. **H. Si**, X. Guo, N. Ansari, C. Chen, L. Duan, J. Huang, "Environment-Aware Positioning by Leveraging Unlabeled Crowdsourcing Data," *IEEE Internet of Things Journal*, vol. 11, no. 9, pp. 16436–16449, 2024.
8. **H. Si**, X. Guo, N. Ansari, "Multi-Agent Interactive Localization: A Positive Transfer Learning Perspective," *IEEE Transactions on Cognitive Communications and Networking*, vol. 10, no. 2, pp. 553–566, 2024.
9. **H. Si**, X. Shao, W. Zhang, "Fuzzy Rule-Based Neural Appointed-Time Control for Uncertain Nonlinear Systems with Aperiodic Samplings," *Expert Systems with Applications*, vol. 170, p. 114504, 2021.
10. **H. Si**, X. Shao, W. Zhang, "MLP-Based Neural Guaranteed Performance Control for MEMS Gyroscope with Logarithmic Quantizer," *IEEE Access*, vol. 8, pp. 38596–38605, 2020.

Co-Author

1. X. Liu, X. Guo, G. O. Boateng, **H. Si**, et al., "A Self-Organizing Federated Reinforcement Learning Framework for Cross-Scenario Collaborative AVP Trajectory Planning," *IEEE Transactions on Automation Science and Engineering*, 2026.
2. X. Liu, X. Guo, G. O. Boateng, **H. Si**, Y. Wang, "Critical Semantic Innovation Interaction Mechanism Towards Vehicle–Infrastructure Collaboration in Autonomous Valet Parking," *IEEE Wireless Communications*, Early Access, pp. 1–9, 2026. doi:10.1109/MWC.2026.3729547.
3. X. Liu, Y. Wang, X. Guo, G. O. Boateng, Y. Cao, **H. Si**, X. Guo, N. Ansari, "ParkingTwin: Training-free streaming 3D reconstruction for parking-lot digital twins," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 240, pp. 114–129, 2026.
4. Y. Wang, X. Liu, X. Guo, **H. Si**, N. Ansari, "O-VIP: A High-Level Semantic Map Construction Method via OSM-Visual-Inertial Fusion for Multivehicle Cooperative AVP," *IEEE Internet of Things Journal*, vol. 13, no. 17, pp. 40592–40608, 2026.

5. Y. F. Aikoun, G. O. Boateng, Z. Wang, **H. Si**, X. Guo, N. Ansari, "Clutter-Aware Waveform Design for Multi-Cell Integrated Sensing and Communication Systems," *IEEE Transactions on Wireless Communications*, vol. 25, pp. 21783–21798, 2026.

6. C. Chen, **H. Si**, G. O. Boateng, X. Guo, D. Yao, "Prioritized Multi-Agent Path Finding for AVPC Leveraging Distributional Dueling DQN," *IEEE Open Journal of Intelligent Transportation Systems*, vol. 7, pp. 1809–1820, 2026.

7. Y. Liu, **H. Si**, G. O. Boateng, X. Guo, B. Qian, N. Ansari, "A Self-Supervised Masked Autoencoder Leveraging Temporal-Frequency Representation for CSI Localization," *IEEE Transactions on Network Science and Engineering*, vol. 13, pp. 7265–7282, 2026.

8. G. O. Boateng, H. Xia, **H. Si**, X. Guo, C. Chen, N. Ansari, "A Platform-Centric Framework for Intelligent Parking Traffic Prediction and Resource Optimization in Shared AVPC Systems," *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 21621–21634, 2025.

9. Y. Cao, X. Guo, G. O. Boateng, **H. Si**, B. Qian, X. Liu, N. Ansari, "ToF-Based Indoor Localization for Harsh Environments: A Continuous Environment Sensing Framework," *IEEE Sensors Journal*, vol. 25, no. 20, pp. 38627–38638, 2025.

10. X. Liu, **H. Si**, G. O. Boateng, X. Guo, Y. Cao, B. Qian, N. Ansari, "Multi-Vehicle Collaborative Trajectory Planning for AVP in Parking Lots: A Bio-Inspired Evolutionary Reinforcement Learning Approach," *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 12, pp. 22789–22803, 2025.

11. T. G. Hailu, X. Guo, **H. Si**, "Indoor Positioning Systems as Critical Infrastructure: An Assessment for Enhanced Location-Based Services," *Sensors*, vol. 25, no. 16, p. 4914, 2025.

12. Y. Cao, X. Guo, G. O. Boateng, N. Ansari, **H. Si**, B. Qian, X. Liu, H. Xia, Y. Liu, "OSM2Net: A Robust Road Network Extraction Framework from Noisy Indoor Parking OpenStreetMap," *IEEE Internet of Things Journal*, vol. 12, no. 15, pp. 30049–30062, 2025.

13. Y. Liu, **H. Si**, G. O. Boateng, X. Guo, Y. Cao, B. Qian, N. Ansari, "Hard Sample Meta-Learning for CIR NLoS Identification in UWB Positioning," *IEEE Internet of Things Journal*, vol. 12, no. 10, pp. 14136–14149, 2025.

14. X. Guo, G. O. Boateng, **H. Si**, Y. Cao, Y. Qiu, Z. Lai, X. Li, X. Liu, C. Chen, "Automated Valet Parking and Charging: A Novel Collaborative AI-Empowered Architecture," *IEEE Communications Magazine*, vol. 63, no. 1, pp. 131–137, 2025.

15. G. O. Boateng, **H. Si**, H. Xia, X. Guo, C. Chen, I. O. Agyemang, N. Ansari, "Automated Valet Parking and Charging: A Dynamic Pricing and Reservation-Based Framework Leveraging Multi-Agent Reinforcement Learning," *IEEE Transactions on Intelligent Vehicles*, vol. 10, no. 2, pp. 1010–1029, 2025.

16. T. G. Hailu, X. Guo, **H. Si**, L. Li, Y. Zhang, "Theories and Methods for Indoor Positioning Systems: A Comparative Analysis, Challenges, and Prospective Measures," *Sensors*, vol. 24, no. 21, p. 6876, 2024.

17. T. G. Hailu, X. Guo, **H. Si**, L. Li, Y. Zhang, "Ada-LT IP: Functional Discriminant Analysis of Feature Extraction for Adaptive Long-Term Wi-Fi Indoor Localization in Evolving Environments," *Sensors*, vol. 24, no. 17, p. 5665, 2024. doi:10.3390/s24175665.

18. Y. Yang, X. Guo, C. Chen, G. O. Boateng, **H. Si**, B. Qian, L. Duan, "TRAIL: A Three-Step Robust Adversarial Indoor Localization Framework," *IEEE Sensors Journal*, vol. 24, no. 7, pp. 10462–10473, 2024.

19. J. Huang, **H. Si**, X. Guo, K. Zhong, "Co-Occurrence Fingerprint Data-Based Heterogeneous Transfer Learning Framework for Indoor Positioning," *Sensors*, vol. 22, no. 23, p. 9127, 2022.

20. X. Shao, **H. Si**, W. Zhang, "Low-Frequency Learning Quantized Control for MEMS Gyroscopes Accounting for Full-State Constraints," *Engineering Applications of Artificial Intelligence*, vol. 115, p. 104724, 2022.

21. X. Shao, **H. Si**, W. Zhang, "Event-Triggered Neural Intelligent Control for Uncertain Nonlinear Systems with Specified-Time Guaranteed Behaviors," *Neural Computing and Applications*, vol. 33, no. 11, pp. 5771–5791, 2021.

22. X. Shao, **H. Si**, W. Zhang, "Fuzzy Wavelet Neural Control with Improved Prescribed Performance for MEMS Gyroscope Subject to Input Quantization," *Fuzzy Sets and Systems*, vol. 411, pp. 136–154, 2021.

23. X. Shao, Z. Cao, **H. Si**, "Neurodynamic Formation Maneuvering Control with Modified Prescribed Performances for Networked Uncertain Quadrotors," *IEEE Systems Journal*, vol. 15, no. 4, pp. 5255–5266, 2021.

Conference Proceedings

1. Y. F. Aikoun, **H. Si**, G. O. Boateng, X. Liu, X. Guo, N. Ansari, "Precoding and Bi-Static Sensing for Cooperative ISAC," in *Proc. IEEE PIMRC*, 2026.

2. **H. Si**, Z. Wang, X. Guo, J. Zhang, Y. Liu, "DOA Estimation for Tri-Polarized Continuous Aperture Array," in *Proc. IEEE International Conference on Communications (ICC)*, pp. 1–6, 2026.

3. Y. F. Aikoun, G. O. Boateng, Y. Liqiang, **H. Si**, X. Guo, "Cognitive ISAC Waveform Design Via Symbol-Level Precoding," in *Proc. 22nd International Computer Conference on Wavelet Active Media Technology and Information Processing (ICCWAMTIP)*, pp. 1–6, 2025.

4. X. Liu, X. Guo, G. O. Boateng, **H. Si**, B. Qian, Y. Cao, H. Xia, Y. Liu, Y. Wang, "A Multi-Vehicle Cooperative Localization Method for AVP Leveraging Double Deep Q-Network," in *Proc. IEEE 11th World Forum on Internet of Things (WF-IoT)*, pp. 1–6, 2025.

5. H. Xia, G. O. Boateng, **H. Si**, X. Guo, B. Qian, X. Liu, Y. Cao, Y. Liu, "Weakly-Supervised Crowdsourcing Localization Based on Self-Learning Spatial-Temporal Attention Network," in *Proc. IEEE/CIC International Conference on Communications in China Workshops (ICCC Workshops)*, pp. 1–6, 2025.

6. K. Chen, G. O. Boateng, **H. Si**, X. Guo, N. Ansari, "A Contract-Based Incentive Mechanism for Optimal Pricing and Allocation in Shared AVPC," in *Proc. IEEE INFOCOM Workshops*, pp. 1–6, 2025.

Preprints (Not Peer-Reviewed)

1. H. G. Tesfay, X. Guo, **H. Si**, "DynaFIPS: Multidimensional Metrics Framework with Confidence-Weighted Fusion for Dynamic Fingerprint-Based Indoor Localization," *Preprints.org*, Aug. 6, 2025. doi:10.20944/preprints202508.0422.v1.

MONOGRAPH

1. X. Guo, H. Li, **H. Si**, L. Li, *Knowledge Transfer and Fusion Theories and Applications for Indoor Positioning.* National Defence Industry Press, Beijing, 2025.

INTELLECTUAL PROPERTY

Granted Chinese Invention Patents

1. X. Guo, L. Chen, **H. Si**, Z. Lai, L. Duan, Q. Dai, Y. Yang, "A Model-Agnostic Meta-Learning Method for Indoor Localization," China Invention Patent, CN116702887B, Aug. 26, 2025.

2. X. Guo, **H. Si**, K. Zhong, J. Huang, Y. Zhang, L. Li, B. Qian, "A Localization and Tracking Method for Indoor Uncertain Systems," China Invention Patent, CN116047896B, Jun. 13, 2025.

3. X. Guo, H. Huang, **H. Si**, L. Han, M. Zhang, B. Qian, J. Huang, Y. Zhang, L. Li, "A Three-Dimensional Localization Method for Unknown Interference in NLoS Environments," China Invention Patent, CN114979951B, Feb. 11, 2025.

4. X. Guo, M. Zhang, **H. Si**, L. Han, H. Huang, B. Qian, J. Huang, Y. Zhang, L. Li, "A Fingerprint-Based Indoor Localization Method Using PPO Reinforcement Learning," China Invention Patent, CN114979952B, Dec. 20, 2024.

5. X. Guo, Y. Zhang, L. Li, **H. Si**, B. Qian, K. Zhong, J. Huang, "A Few-Shot Radar Target Recognition Method Based on Hierarchical Meta-Transfer Learning," China Invention Patent, CN115345322B, Feb. 7, 2023.

6. X. Shao, **H. Si**, W. Zhang, J. Liu, "A Reliable Attitude Control Method for Quadrotors Considering Onboard Gyroscope Failures," China Invention Patent, CN111338371B, Aug. 23, 2022.

GRANTS & SCHOLARSHIPS

External

National Graduate Scholarship (State Council of China) 2025
China Scholarship Council (CSC) Scholarship 2024
Tencent Scholarship (Special Class / Top Award) 2024
Outstanding Graduate of Sichuan Province (Dept. of Education) 2025

Internal (UESTC)

First-Class Academic Scholarship 2024–2025
Academic Rising Star 2023
Outstanding Graduate Student 2023–2025

SERVICE & PROFESSIONAL DEVELOPMENT

Professional Service

- **Awardee, CAST Young Talent Support Program (PhD Special Pilot)** 2024–2026
  - Participation in academic community engagement, conference activities, paper development support, and domestic/international exchanges (program support up to two years).
- **Journal Reviewer (Ad Hoc / Regular)**
  - **IEEE**: TSP; TWC; TMC; TCOM; TCCN; IoT-J; WCL.
  - **Elsevier**: Internet of Things; Expert Systems with Applications.
  - **Nature Portfolio**: Scientific Reports.
- **Membership**: IEEE Communications Society (Student Member).

Teaching & Instructional Service

- **Teaching Assistant**, *Innovation Thinking and Invention Methods* 2021–2022
  - Assisted course delivery; supported in-class facilitation and student engagement.
  - Graded assignments/projects and maintained consistent assessment records.
  - Organized course materials (lecture notes and learning progress documentation) to support course continuity.
- **Undergraduate Thesis/Capstone Supervisor (B.Eng.)**, UESTC 2023–2024
  - Supervised two undergraduate capstone projects.
  - Topics: (1) GNN-based wireless localization; (2) DRL-based mapless robot navigation.
  - Provided milestone planning, technical mentoring, and thesis-writing guidance.

TECHNICAL SKILLS

- **Programming**: MATLAB, Python, PyTorch.
- **Tools**: Linux, FEKO, Blender.
- **Research**: simulation/modeling; ML training and evaluation; localization algorithm design.